

DO NOT WRITE ON TEST



**FINANCIAL MATH &
ANALYSIS CONCEPTS**

~OPEN EVENT~

(190)

STATE 2021

TOTAL POINTS

_____100 Points

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts if any.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.
4. For any interest calculations, assume a 360-day year.
5. Unless otherwise stated, assume that all plant assets were purchased on January 1.
6. Unless otherwise stated, assume that the overtime rate is 1.5 regular pay for any hours worked in excess of 40 hours per week.

Identify the letter of the choice that best completes the statement or answers the question.

1. Cora hopes to save \$1,200 to pay for her textbooks next semester. If Cora earns \$1,040 each month, how much should she save each month if she has five months to accomplish her goal?
 - A. \$240
 - B. \$208
 - C. \$160
 - D. \$140

2. Digital Solutions estimates that sales returns will be 6% of sales next year. Sales for the year were \$1,436,509. How much is Digital Solutions expecting in sales returns?
 - A. \$90,524.56
 - B. \$134,569.25
 - C. \$430,953.70
 - D. \$86,190.54

3. Matthew pays \$240 a month in health insurance. His policy has a \$5,000 deductible and 20% coinsurance. Matthew has had three treatments on his back, and has paid \$1,300 each time. The treatments have *not* been effective, and Matthew needs to have surgery. How much will he have to pay towards his deductible before his co-insurance kicks in?
 - A. \$2,120
 - B. \$1,100
 - C. \$1,300
 - D. \$1,000

4. Robert is a customer of Digital Solutions. Robert has requested more than the 30 days that a normal Accounts Receivable would give him. Both parties agreed on a \$9,600, 60-day note that was signed on August 12. If Robert paid \$9,776 on the maturity date, what rate of interest did Digital Solutions charge Robert?
 - A. 9%
 - B. 16%
 - C. 11%
 - D. 29%

5. A box of Tom Tiger cereal costs \$3.25 at the grocery store. The store's app has a 5% off coupon, and Rose has a paper coupon for \$0.25 off. How much will a box of Tom Tiger cost after the coupons have been applied?
 - A. \$3.00
 - B. \$2.85
 - C. \$2.84
 - D. \$1.88

6. A smartboard is purchased on July 1 for \$9,400. It has a useful life of five years and a salvage value of \$1,100. How much depreciation will be taken in the first year under the straight-line method?
- A. \$1,660
 - B. \$1,880
 - C. \$830
 - D. \$2,100
7. If a plant asset costs \$4,200, has accumulated depreciation of \$2,600, and is sold for \$2,500, the gain or loss on disposal is _____.
- A. \$3,500 gain
 - B. \$3,500 loss
 - C. \$900 gain
 - D. \$900 loss
8. The regular price of an ounce of correction fluid is \$1.84. A 12-ounce bottle of correction fluid can be purchased for \$17.04. How much would be saved by purchasing correction fluid by the 12-ounce bottle?
- A. \$1.84
 - B. \$5.04
 - C. \$11.08
 - D. There is no price differential.
9. Carson is a website designer for Grantham Inc. Carson earns a regular hourly rate of \$26.12, as well as time and a half for overtime work on Saturdays and double time on Sundays. This week Carson worked 40 hours from Monday through Friday, 4 hours on Saturday, and 4 hours on Sunday. What is Carson's total pay for the week?
- A. \$1,201.52
 - B. \$1,149.28
 - C. \$1,044.80
 - D. \$1,410.48

10. Pattmore's Produce purchased 2,000 pounds of potatoes at 34 cents/pound and is shipping them from Texas to his store in Baltimore. During the trip, Pattmore anticipates 60 pounds of shrinkage by the time it reaches the loading dock. From the dock to the time sitting on the shelves, Pattmore anticipates 10% of shrinkage from the remaining potatoes.
- With total shrinkage allocated, what would be the total loss in dollars from the original purchase?
- A. \$65.96
 - B. \$68.00
 - C. \$86.36
 - D. \$88.40
11. Downton Co. sells annual magazine subscriptions for \$66. If Mary paid for her annual subscription in full on August 1, 2020, how much revenue will Downton Co. record for Mary's subscription in 2020?
- A. \$33.00
 - B. \$27.50
 - C. \$0.00
 - D. \$66.00
12. Daisy earns a weekly salary of \$1,272 at the Grantham Arms. Next month Daisy will be promoted from Assistant Manager to Senior Manager. In her new position, she will be paid \$3,594 bi-weekly. How much more per year will Daisy earn as a Senior Manager than as an Assistant Manager?
- A. \$20,112
 - B. \$13,112
 - C. \$13,867
 - D. \$27,300
13. The Raw Materials account had a balance of \$835 at the beginning of the quarter. During the quarter, \$1,324 of materials were purchased, and \$2,012 of materials were used in production. What is the balance in Raw Materials at the end of the quarter?
- A. \$688
 - B. \$2,501
 - C. \$1,523
 - D. \$147
14. Branson Mfg. purchased materials on account from Leech Inc. for \$4,800 on December 10. Leech offered Branson credit terms of 2/10, n/30. If Branson pays Leech on December 22, how much does Branson owe?
- A. \$4,800
 - B. \$4,320
 - C. \$4,704
 - D. \$3,360

15. Alicia is traveling to Paris and needs to convert her US dollars to local currency. If the conversion rate is \$1.10, how many euros can Alicia get for \$500?
- A. €500.00
 - B. €550.00
 - C. €454.55
 - D. €505.00
16. Digital Solutions grants all employees a 3% pay raise. If Henry currently earns \$2,200 semi-monthly, how much will his salary increase by annually after the pay raise?
- A. \$792
 - B. \$1,584
 - C. \$7,920
 - D. \$2,134
17. William has \$56 in his checking account on Monday morning. William's paycheck of \$1,900 was deposited on Tuesday morning; and on Tuesday afternoon, William used his debit card to buy groceries totaling \$87. On Wednesday William used his debit card to fill up his car with gas for \$23. On Thursday William paid three bills – rent \$500, electric bill \$175, and car insurance \$125. If William intends to save 5% of his paycheck, how much will he have left over at the end of the week?
- A. \$951
 - B. \$1,046
 - C. \$895
 - D. \$996
18. Edith's employer allots each employee \$1,800 a year for travel, and any amounts spent over this budget will *not* be reimbursed by the company. Edith recently traveled to a professional conference and spent \$300 on conference registration, \$600 on hotel, \$355 on flight, \$192 on meals, \$36 on parking at the airport, \$68 on transportation at her destination, and \$60 in checked-bag fees. If Edith wants to attend a second conference this year, what is the maximum amount that her employer will contribute?
- A. \$1,800
 - B. \$189
 - C. \$1,611
 - D. All expenses will be reimbursed.

19. Branson has reserved a hotel room for four nights. The hotel rate is \$199 per night, with a 15% hotel tax rate, a 1% tourism tax rate, and a 3% business district tax rate. Parking is charged at a rate of \$20 per calendar day, and Internet is \$15 per night. How much will Branson's hotel bill be in total?
- A. \$1,113.84
 - B. \$1,107.24
 - C. \$1,359.58
 - D. \$947.24
20. Sybill normally works 40 hours a week and earns \$680. This week Sybill worked 48 hours. How much will Sybill earn this week?
- A. \$850
 - B. \$884
 - C. \$816
 - D. \$680
21. Nicholas purchased a piece of property with $\frac{1}{2}$ acre of land and an office building for \$950,000. When appraised separately, the land appraised at \$300,000 and the office building appraised at \$700,000. Using the relative-market value method, at what amount will Nicholas record the land at in his company records?
- A. \$285,000
 - B. \$158,500
 - C. \$159,250
 - D. \$155,750
22. Digital Solutions remitted 1,631.40 to the state government for weekly sales tax. If the state sales tax rate is 7.65%, how much were Digital Solutions' sales for the week?
- A. \$21,480.21
 - B. \$21,325.49
 - C. \$12,664.59
 - D. \$18,395.43
23. Maggie owns an accounting company. During the month of November, Maggie spent 173 hours working on jobs for customers. The job for Mr. Smith took 43.25 hours to complete and \$545 in materials. Maggie pays her employees \$13 an hour. Utilities for the month of November were \$849.43, and Maggie assigns utilities to each job based on the number of labor hours worked. How much did it cost Maggie to complete the job for Mr. Smith?
- A. \$1,091.00
 - B. \$206.22
 - C. \$546.00
 - D. \$1,319.61

24. Jamie buys a new car with a sticker price of \$17,999. Jamie made a down payment of \$1,000 and agrees to make monthly payments of \$335 for 5 years. How much will Jamie have actually paid for the car by the time she finishes paying for it?
- A. \$20,100
 - B. \$18,999
 - C. \$19,100
 - D. \$21,100
25. Knox is setting up a budget for his family of 4. If Knox earns \$940 each week and spends \$120 each week on groceries to feed his family for 7 breakfasts, 7 dinners, and 2 lunches, what percentage of his income is spent on food?
- A. 25%
 - B. 13%
 - C. 30%
 - D. 22%
26. Charles' average electric bill is \$190 a month. Charles wants to put in a pool, but is told to expect his electric bill to increase by 10% on average. If Charles puts in a pool, how much more should Charles expect to pay for electricity in a year?
- A. \$1,900
 - B. \$19
 - C. \$228
 - D. \$418
27. Devonte purchased a new laptop for school for \$1,136.00. His credit card is offering 18 months interest-free financing. How much should Devonte pay each month to make sure that the laptop is paid off before the interest-free period expires?
- A. \$94.67
 - B. \$75.73
 - C. \$56.80
 - D. \$63.11
28. Elsie's favorite store is offering 30% off all merchandise. Elsie also has a coupon for 20% off any one item. If Elsie buys a purse with a regular price of \$56, how much will she pay for the purse after the discounts?
- A. \$31.36
 - B. \$39.20
 - C. \$44.80
 - D. \$28.00

29. Clint is hosting a training workshop. Clint has rented a room that can hold 25 people for \$300. Clint will provide handouts to each participant, which cost \$4.80 each to copy, as well as a combination pen and highlighter, which each cost \$0.36. He estimates that snacks will cost a total of \$50. If Clint sells tickets to the workshop for \$25 each, how many tickets does Clint need to sell before he will make a profit?
- A. 23 tickets
 - B. 18 tickets
 - C. 12 tickets
 - D. 17 tickets
30. Digital Solutions is planning to replace the carpet in their conference room. If the room measures 18 feet long by 22 feet wide with 10 feet tall ceilings and the carpet costs \$1.22/square foot, what will be the total cost of the carpet?
- A. \$589.41
 - B. \$483.12
 - C. \$396.00
 - D. \$97.60
31. Tolbert purchased 25 shares of Digital Solutions' stock when the stock was trading at \$126 per share. If Tolbert sold the stock at \$131.48, how much did he make off the sale?
- A. \$137.00
 - B. \$25.00
 - C. \$19.52
 - D. \$304.00
32. Digital Solutions declared a \$500,000 dividend. If Digital Solutions has 15,000 shares of 6% \$100 par preferred stock and 135,000 shares of \$1 par common stock outstanding, how much will Digital Solutions pay out per share of common stock?
- A. \$3.04
 - B. \$3.70
 - C. \$3.33
 - D. \$0.22
33. Denise, Anthony, and Rudy are all in a partnership together. In the partnership agreement, they agreed to a 3/3/2 split, respectively, of all partnership profits and losses, after owner withdrawals. Last year the partnership made \$138,000 in profits, but Denise withdrew \$25,000, Anthony withdrew \$30,000, and Rudy withdrew \$18,000. How much of the partnership profits will be distributed to Rudy?
- A. \$17,250
 - B. \$16,500
 - C. \$16,250
 - D. \$34,500

34. Albert purchased a boat seven years ago for \$96,000. Albert sold the boat today for \$42,000 and recorded a loss of \$12,300. What was accumulated depreciation on Albert's boat?
- A. \$41,700
 - B. \$54,000
 - C. \$29,700
 - D. \$54,300
35. Last year Digital Solutions had taxable profit of \$1,386,000. If Digital Solutions paid \$485,100 in taxes, what was their corporate tax rate?
- A. 30%
 - B. 25%
 - C. 35%
 - D. 29%
36. Gedeon has earned scholarships to pay for 82% of her tuition for the semester. If Gedeon had to pay \$972 out-of-pocket for tuition, what was the total cost of her tuition?
- A. \$4,428
 - B. \$7,970
 - C. \$1,185
 - D. \$5,400
37. Spencer is a professional baseball player and is negotiating a contract with his new team. Spencer is asking for \$800,000 for the year with an additional \$6,000 for every game he starts. His new team has offered him \$10,000 for every game he starts, but only \$700,000 for a base salary. How many games would he need to start in order to make more with the team's offer?
- A. 15 games
 - B. 25 games
 - C. 26 games
 - D. 40 games
38. Molly is searching for a basic cell phone voice plan for her grandmother. She is interested in being able to have a cell phone for limited use and in case of emergencies. Molly has found two possible plans:
- Plan A: \$29 fee per month plus \$0.05 for each minute
 - Plan B: \$10 fee per month plus \$0.25 for each minute

If Molly's grandmother usually talks on the phone for two hours each month, which plan would be the better deal?

- A. Plan A
- B. Plan B
- C. Both plans would be the same.
- D. It is impossible to determine which plan is better.

39. Last month Betty worked 42 hours the first week, 40 hours the second week, 38 hours during week three, and 44 hours in week four. If Betty's gross pay for those four weeks was \$3,006, what was Betty's hourly rate?
- A. \$18.00
 - B. \$18.11
 - C. \$18.33
 - D. \$18.79
40. The owner of Crawley Inc. has narrowed the purchase of a used delivery van to two choices. Both choices (A & B) have the same list price of \$24,000. Van A is available with an 8% discount and \$900 rebate. Van B is advertised for 15% off. Which van purchase will end up costing less and by how much?
- A. Van A: \$ 780
 - B. Van B: \$ 780
 - C. Van A: \$2,864
 - D. Van B: \$2,864
41. Clermont Co. pays its employees semi-monthly. Clermont offers the employees a fringe benefit package that includes accumulating hours for the following: 2 hours of vacation per pay period; 1 hour of sick time per pay period; and .5 hours of personal leave per pay period. What would be the total amount of hours earned per year within the fringe benefit package?
- A. 196
 - B. 42
 - C. 91
 - D. 84
42. A new laser printer can be purchased for \$800 cash, or 25% down with 12 monthly payments of \$55 each. How much money is saved by paying cash?
- A. \$30
 - B. \$40
 - C. \$60
 - D. \$75
43. As a salesman for McCoy Co., Rod receives \$.34 per mile travel reimbursement. Rod traveled 300 miles on March 14, 250 miles on March 19, 175 miles on March 26, and 400 miles on March 28. How much will Rod be reimbursed for March mileage?
- A. \$190.25
 - B. \$245.38
 - C. \$382.50
 - D. \$201.48

44. The total sales for Diana Craft and Candle Company last year were \$158,946.23. It was determined that 32% of sales were made in Europe. What amount of sales were made within the United States?
- A. \$50,862.79
 - B. \$108,083.44
 - C. \$108,095.68
 - D. \$55,637.51
45. The bookkeeper deposited these items in the bank: 13 twenty dollar bills, 24 quarters, 14 dimes; and two checks of \$124.27 and \$187.65. She received \$50 back in cash. What was the net amount of her deposit?
- A. \$429.92
 - B. \$479.92
 - C. \$529.32
 - D. \$579.32
46. Daniel has a credit card balance of \$1,400. Payments made to the card company are applied to interest first and the remainder to the balance, with the interest calculated on the balance owed. The credit card carries an annual rate of 23%. If the first monthly payment is \$25, how much will go toward the balance owed after the interest is deducted from the payment?
- A. \$26.83
 - B. \$1.83
 - C. \$3.22
 - D. \$0.00
47. Rachel buys 100 shares of stock at \$93 per share. The price goes up by 12%, and she sells 50 shares. How much did she sell the 50 shares for?
- A. \$104.16
 - B. \$112.00
 - C. \$11.16
 - D. \$81.84
48. Harris Mfg. uses a profit-sharing plan for employees' retirement. Ninety employees equally share the company's year 2020 profit of \$325,350. What amount will each employee receive?
- A. \$4,098.17
 - B. \$3,525.30
 - C. \$3,915.00
 - D. \$3,615.00

49. Gene is paid \$7.12 per hour with time-and-a-half overtime for any hours over eight per day plus a commission. His commission consists of 2.5% of the first \$3000 in sales and 5% on all sales over \$3000. Total sales for the week were \$3,977. Find his gross pay for the week he worked 8 hours on both Monday and Tuesday, 7 hours on Wednesday, 6 hours on Thursday, 4 hours on Friday, and 7 hours on Saturday.
- A. \$284.80
 - B. \$408.65
 - C. \$123.85
 - D. \$483.65
50. Clark earns \$15 per hour and is paid bi-weekly. If Clark works 38 hours one week and 42 hours the next week, what are his total bi-weekly earnings?
- A. \$1,230.00
 - B. \$1,200.00
 - C. \$1,215.00
 - D. \$1,245.00